Problem 5.10

Henry invests \$1000 on January 15 in an account earning simple interest at an annual effective rate of 10%. On November 25 of the same year, Henry withdraws all his money. How much money will Henry withdraw if the bank counts days:

(a) using exact simple interest (ignoring February 29th),

(b) using ordinary simple interest,

(c) using Banker's Rule.

We use the simple interest formula:

$$I = Prt,$$

where $P = 1000$, $r = 0.10$, and t is the time in years.

1. Number of Days

From January 15 to November 25, the exact number of days is:

$$16 + 28 + 31 + 30 + 31 + 30 + 31 + 31 + 30 + 31 + 25 = 314 \text{ days.}$$

For ordinary simple interest, 30-day months are used. From January 15 to November 25:

$$15 + (30 \cdot 9) + 25 = 310 \text{ days.}$$

(a) Exact Simple Interest

Using the exact number of days and a 365-day year:

$$I = 1000(0.10)\left(\frac{314}{365}\right).$$

Therefore,

$$I \approx 86.03.$$

The total amount withdrawn is

$$1000 + 86.03 = 1086.03.$$

(b) Ordinary Simple Interest

Using 310 days and a 360-day year:

$$I = 1000(0.10)\left(\frac{310}{360}\right).$$

Thus,

$$I \approx 86.11.$$

The total amount withdrawn is

$$1000 + 86.11 = 1086.11.$$

(c) Banker's Rule

Using the exact 314 days and a 360-day year:

$$I = 1000(0.10)\left(\frac{314}{360}\right).$$

So,

$$I \approx 87.22.$$

The total amount withdrawn is

$$1000 + 87.22 = 1087.22.$$